SPHINGOLIPIDS DRIVE SURFACTIN-INDUCED PLANT IMMUNITY THROUGH MEMBRANE REMODELLING

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ABSTRACT

Plant-associated beneficial rhizobacteria are able to stimulate and enhance plant immunity through the secretion of metabolites, such as cyclic lipopeptides (CLPs). While the plant disease reduction due to CLPs is well-documented, their sensing mechanisms at the plant cell surface remains largely unresolved.

Combining biological and biophysical approaches, we demonstrated that the immune activation in *Arabidospsis thaliana* is mediated by a specific interaction with membrane sphingolipids, leading to changes in membrane mechanical properties and structure. These physical changes are then perceived by mechanosensitive channels which activate ion fluxes and initiate early immune signaling. This mechanism results in systemic defense potentiation and enhanced resistance to the necrotrophic fungus *Botrytis cinerea*.

Our findings uncover a lipid-mediated sensing mechanism for bacterial CLPs by plant cells that is distinct from the classical description of plant immunity, relying on protein pattern recognition receptors to initiate pattern triggered immunity. Beyond, its fundamental significance, understanding the chemical dialogue between rhizobacteria and host plant is a promising alternative to chemical crop protection strategies.

REFERENCES

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